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[EXAMPLE] Adaptive wall modelling approach in Large Eddy Simulation with the Immersed Boundary Method.

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Chapter 1

Introduction

1.1 Computational Fluid Dynamics

Computational Fluid Dynamics (CFD) is the study of fluids through a numerical computer-based approach, using algorithms to find solutions to the equations that describe the fluid behavior. The information produced have mainly two purposes: in the field of research, it allows a better understanding of the physical mechanisms governing the fluid motion, and the consequent modelling and control of such mechanisms. In the engineering analyses, fluid simulation is used as a prediction tool, especially during the product design phase, to assess flow characteristics and physical parameters, such as lift and drag forces generated, propagation of acoustic waves or convective heat transfer.

As all mathematical models, the definition of the level of approximation have a primary importance also for fluid simulation. This choice concern various aspects of the description of the phenomena. Firstly the scale of reality considered: the reality can be described at an atomic, particle or micro- and macroscopic level. The last one is that used by classical mechanics, in particular continuum mechanics, that is the case considered in CFD.

The second aspect to be defined in terms of level of approximation is the accuracy of dynamic description. It consist on the choice of which forces should be considered exerted on the system, based on their relevance. The most general case is represented by the Navier-Stokes equations complemented by constitutive laws (empirical laws characterizing the fluid and state laws). This case can be simplified, obtaining for example the Euler equations, for the description of inviscid flows.

Another aspect regards the scale separation problem, in particular the spatial and temporal resolution of choice. To obtain a description of maximum quality, all the space-time scales of the solution must be taken into account while solving the unsteady Navier-Stokes

equations. As will be discussed later, it turns out that such a solution criterion is extremely restrictive for multi-scale phenomena, like in the case of turbulent flows.

1.2 Turbulent flows

In most of the engineering applications, flows are prevalently turbulent. Generally this is the case of fluid processed by pumps, compressors or flows around cars, trains, airplanes and many more situations in the everyday surrounding.

Such flows are characterized by some specific properties. A key feature is the multi-scale nature of the vortical structures, whose dimensions span 6 to 8 orders of magnitude. Despite the deterministic nature (they can be described accurately by the Navier-Stokes equations), the strong dependence and sensitivity to initial conditions make their behavior chaotic. Furthermore, turbulent flows are intrinsically three-dimensional and unsteady, they show an high diffusivity (mixing of momentum, energy and species) and also an increased wall friction when compared to laminar flows.

The flow regime is described by the Reynolds number, a non-dimensional value representing the ratio between inertial and viscous forces, and defined as:

$$Re = \frac{\rho\, U_0\, L_0}{\mu} = \frac{U_0\, L_0}{\nu} \tag{1.1}$$

where U_0 and L_0 are the characteristic velocity and length scale of the flow. It is observed that, for higher values of Re, the magnitude of velocity gradients is progressively more intense, and the scale of smaller vortices decreases.

Considering the chaotic nature of the turbulence, a deterministic and instantaneous description of the phenomena is not always the most efficient option. Consequently, the analysis is typically focused on a statistical approach, considering averaged fields. It can be observed how those averaged fields resemble qualitatively the laminar behavior, and their dependence on the Reynolds number diminishes significantly.

1.2.1 Reynolds-Averaged Navier-Stokes equations

To analytically formalize the idea of considering the average fields and the turbulent one separately, the Reynolds averaging operator and the Reynolds decomposition are here introduced. The former consist in a ensemble average over a number of realizations. In more practical terms this implies repeating the same experiment N times, under identical conditions, and averaging the flow variables at the exact same location and time instant across all the independent measures. Considering to apply the Reynolds operator for the

velocity component u:

$$\langle u(\boldsymbol{x},t)\rangle = U(\boldsymbol{x},t) = \frac{1}{N}\sum_{k=1}^{N\gg 1} u_{(k)}(\boldsymbol{x},t) \tag{1.2}$$

Instead for what concerns the Reynolds decomposition, it can be defined as:

$$u(\boldsymbol{x},t) = U(\boldsymbol{x},t) + u'(\boldsymbol{x},t) \tag{1.3}$$

where u' is the fluctuation component and U is the averaged quantity.

By substituting in the Navier-Stokes equations the Reynolds decomposition and applying the Reynolds operator the Reynolds-Averaged Navier-Stokes (RANS) equations are obtained:

$$\begin{cases} \boldsymbol{\nabla}\cdot\boldsymbol{U} = 0 \\ \dfrac{D\boldsymbol{U}}{Dt} = -\dfrac{\boldsymbol{\nabla}P}{\rho} + \boldsymbol{\nabla}\cdot(2\nu\boldsymbol{E}) - \boldsymbol{\nabla}\cdot\langle\boldsymbol{u}'\boldsymbol{u}'\rangle \end{cases} \tag{1.4}$$

where the term $\langle\boldsymbol{u}'\boldsymbol{u}'\rangle$ is the Reynolds stress tensor, and its modelling represents the closure problem of the RANS equations.

1.2.2 Energy cascade

The RANS equations introduced above, allow demonstrating some fundamental mechanisms governing the energy dynamics within turbulent flows. Omitting for brevity the detailed algebraic steps, the mean and turbulent flow kinetic energies can be defined respectively as:

$$\mathcal{K} = \frac{1}{2}\boldsymbol{U}\cdot\boldsymbol{U} = \frac{1}{2}U_i U_i \qquad k = \frac{1}{2}\langle\boldsymbol{u}'\cdot\boldsymbol{u}'\rangle = \frac{1}{2}\langle u_i' u_i'\rangle \tag{1.5}$$

By manipulating the momentum equation, it is possible to derive a formulation for both the time derivative of the mean kinetic energy and turbulent kinetic energy terms:

$$\frac{D\mathcal{K}}{Dt} = \underbrace{\frac{\partial}{\partial x_j}\left[-\frac{P}{\rho}U_j + 2\nu\, E_{ij}\, U_i - \langle u_i'u_j'\rangle\, U_i\right]}_{\boldsymbol{\nabla}\cdot\boldsymbol{F}_{\mathcal{K}}} \underbrace{-2\nu\, E_{ij}E_{ij}}_{-\mathcal{D}} + \underbrace{\left(\langle u_i'u_j'\rangle\frac{\partial U_i}{\partial x_j}\right)}_{-\mathcal{P}} \tag{1.6}$$

$$\frac{Dk}{Dt} = \underbrace{\frac{\partial}{\partial x_j}\left[-\frac{\langle p'u_j'\rangle}{\rho} + 2\nu\,\langle e_{ij}'u_i'\rangle - \frac{\langle u_i'u_i'u_j'\rangle}{2}\right]}_{\boldsymbol{\nabla}\cdot\boldsymbol{F}_k} \underbrace{-2\nu\,\langle e_{ij}'e_{ij}'\rangle}_{-\mathcal{E}} + \underbrace{\left(-\langle u_i'u_j'\rangle\frac{\partial U_i}{\partial x_j}\right)}_{\mathcal{P}} \tag{1.7}$$

where $\boldsymbol{F}_{\mathcal{K}}$ and $\boldsymbol{F}_k$ are the fluxes of kinetic energy (spatial transport terms), $\mathcal{D}$ and $\mathcal{E}$ are the dissipation terms respectively for the mean and turbulent flows and $\mathcal{P}$ is the turbulent

energy production. In a more compact form, the two equation can be written as:

$$\frac{D\mathcal{K}}{Dt} = \boldsymbol{\nabla} \cdot \boldsymbol{F}_{\mathcal{K}} - \mathcal{D} - \mathcal{P} \qquad \frac{Dk}{Dt} = \boldsymbol{\nabla} \cdot \boldsymbol{F}_k - \mathcal{E} + \mathcal{P} \tag{1.8}$$

Following the approach described by Pope [1], the equations can be integrated over a sufficiently large control volume Ω, surrounding a free shear flow such as a turbulent jet ($Re \gg 1$), in stationary conditions. This perspective, and what result from the integration, allows to identify which component are the dominant in the energy dynamics.

$$\int_\Omega \frac{\partial \mathcal{K}}{\partial t}\, d\Omega = \mathbb{C}_{\mathcal{K}} + \mathbb{F}_{\mathcal{K}} - \mathbb{D}_{\mathcal{K}} - \mathbb{P} \implies \mathbb{C}_{\mathcal{K}} \approx \mathbb{P} \tag{1.9}$$

$$\int_\Omega \frac{\partial k}{\partial t}\, d\Omega = \mathbb{C}_k + \mathbb{F}_k - \mathcal{E} + \mathbb{P} \implies \mathbb{P} \approx \mathcal{E} \tag{1.10}$$

The analysis of the dominant terms in the energy balance equations reveals how the input of energy, associated with convection ($\mathbb{C}_{\mathcal{K}}$) and its viscous dissipation ($\mathcal{E}$) occur at different scales. The apparent decoupling between energy input and output is better described by Richardson's theory.

1.2.3 Richardson's theory

The first concept in Richardson's view of the energy cascade is that the turbulence can be considered to be composed of eddies of different sizes [1].

Considering the eddies with the largest sizes, their characteristic dimensions can be evaluated as: $\ell \approx \ell_0 \approx L_0$ and $u \approx u_0 \approx u'_{rms} = \sqrt{(2/3) \cdot k} \approx (0.1 - 0.3)\, U_0$ (where the last relation comes from observation), thus their Reynolds number:

$$Re_0 = \frac{u_0\, \ell_0}{\nu} \approx \frac{U_0\, L_0}{\nu} \gg 1 \tag{1.11}$$

this suggests how at these scales the direct effect of molecular viscosity is negligible.

Thus, in order to achieve dissipation, unstable large eddies transfer energy to smaller ones. This process continues until the scale $\ell = \eta$ is reached, at this level the associated Reynolds number becomes sufficiently small, or in other words the molecular viscosity begins to be effective in energy dissipation.

The energy transferred to smaller scales can be evaluated as:

$$\frac{k_0}{\tau_0} \approx \frac{u_0^2}{(\ell_0/u_0)} = \frac{u_0^3}{\ell_0} \propto \frac{U_0^3}{L_0} \propto \mathcal{E} \tag{1.12}$$

the scaling of $\mathcal{E}$ with U_0^3/L_0, independently from ν (at the high Reynolds numbers considered) proves to be consistent with the experimental results for free shear flows [1].

1.2.4 Kolmogorov's theory

With the Kolmogorov theory, a better characterization of the involved scales in high-Reynolds flows was proposed. The theory was stated in three hypotheses:

- **Local isotropy hypothesis:** *At sufficiently high Reynolds numbers, the small-scales ($\ell \ll \ell_0$) turbulent motions are statistically isotropic.*

  While larger eddies show an anisotropic behavior, as a result of flow geometry and boundary conditions, scales of motion under a certain length value (small-scales) have statistical properties that are independent of the large, anisotropic scales. Consequently, they exhibit a universal behavior, in the sense that they appear similar in every high-Reynolds number flow.

- **First similarity hypothesis:** *In every turbulent flow at sufficiently high Reynolds number, the statistics of the small-scale motions have a universal form that is uniquely determined by ν and $\mathcal{E}$.*

  As a consequence of this hypothesis, starting from the ν and $\mathcal{E}$ parameters, it is possible to identify a particular set of scales (Kolmogorov scales) with specific length, velocity and time characteristics. This result is obtained by dimensional analysis.

$$\eta = \left(\frac{\nu^3}{\mathcal{E}}\right)^{1/4} \qquad u_\eta = (\nu\mathcal{E})^{1/4} \qquad \tau_\eta = \left(\frac{\nu}{\mathcal{E}}\right)^{1/2} \tag{1.13}$$

  These scales are considered the smallest active scales in the turbulent motion. As a support for this assertion it is possible to observe that the related Reynolds number is unity ($Re_\eta = \eta \cdot u_\eta/\nu = 1$), meaning that viscous forces effectively dominate over inertial forces, allowing the dissipation phenomena.

  It is possible to evaluate the entity of scales separation, considering the ratios of the smallest to largest scales. These ratios depend on the Reynolds number according to the following relations:

$$\frac{\ell_0}{\eta} \sim Re^{3/4} \qquad \frac{u_0}{u_\eta} \sim Re^{1/4} \qquad \frac{\tau_0}{\tau_\eta} \sim Re^{1/2} \tag{1.14}$$

- **Second similarity hypothesis:** *In every turbulent flow at sufficiently high Reynolds number, the statistics of the motions of scale ℓ in the range $\eta \ll \ell \ll \ell_0$ have a universal form that is uniquely determined by $\mathcal{E}$, independent of ν.*

  This intermediate range of scales is known as the inertial subrange. In this region, the eddies are large enough that viscous effects are negligible (since $\ell \gg \eta$), but small enough that the large-scale anisotropic geometry is not felt (since $\ell \ll \ell_0$).

Consequently, the energy cascade is dominated mainly by inertial forces. Again by dimensional analysis:

$$u(\ell) = (\mathcal{E}\,\ell\,)^{1/3} \qquad \tau(\ell\,) = \left(\frac{\ell^{\,2}}{\mathcal{E}}\right)^{1/3} \qquad Re(\ell\,) = \left(\frac{\ell}{\eta}\right)^{4/3} \tag{1.15}$$

In summary, the energy input associated with the mean velocity field (for example as a result of convection) is in part transferred from one point of the domain to another by the flux term, another part is transferred down to the inertial subrange by the production term. This transfer to smaller and smaller scales continues until the Kolmogorov scales are reached, the effects of viscosity become dominant and the energy can be dissipated as heat.

1.3 Simulation techniques

The theoretical framework just introduced, highlights the wide range of scales of motion involved in the turbulent dynamics. In order to describe properly the fluid flow, it is necessary to take into account in the simulation all the space-time scales affecting the phenomena. This can result in an enormous amount of degrees of freedom, especially for high-Reynolds flows.

1.3.1 Direct Numerical Simulation

The most direct approach to perform a simulation, is choosing a space (grid) and time discretization able to represent all the space-time scales, included the smaller ones (Kolmogorov scales), in a non-stationary, three dimensional domain.

In terms of accuracy and level of description provided it is unrivalled at the point that these numerical experiments can be considered directly analogous to actual laboratory experiments [2] and does not need validation. However for the number of degrees of freedom involved its cost is extremely high and its applicability is limited to low or moderate Reynolds numbers [1].

The scale separation problem can be illustrated assuming to discretize a domain with dimensions $L \sim \ell_0$. To properly resolve the dissipative structures, the grid spacing Δ must be comparable to the Kolmogorov scale η. Applying the relations (1.14), the number of grid points required along each axis is on the order of $Re^{3/4}$ and in terms of time discretization, the required number of time-steps for a τ_0 interval would be on the order of $Re^{1/2}$. Overall, the computational cost of the simulation scales with the Reynolds number, following a $Re^{11/4} \approx Re^3$ relation.

As a consequence, this type of simulation for large Reynolds number, such as those related to application in the aeronautical field ($Re \sim 10^8$) requires computer resources very much greater than currently available super computer capacities, and is therefore not practicable [3].

1.3.2 RANS-based simulation

An approach capable of reducing heavily the number of degrees of freedom to be computed is based on the Reynolds-Averaged Navier Stokes equations (1.4) previously described. In contrast to the direct resolution of all turbulent scales, such method focuses on the determination of the mean flow field. The contribution related to the fluctuation field (transfer of momentum due to turbulent fluctuations) appears in the equations as the Reynolds stress tensor $\langle \boldsymbol{u}'\boldsymbol{u}' \rangle$ that needs to be determined to close the system of equations.

The structure of the Reynolds stress tensor is:

$$\langle \boldsymbol{u}'\boldsymbol{u}' \rangle = \begin{bmatrix} \langle u'^2 \rangle & \langle u'v' \rangle & \langle u'w' \rangle \\ \langle v'u' \rangle & \langle v'^2 \rangle & \langle v'w' \rangle \\ \langle w'u' \rangle & \langle w'v' \rangle & \langle w'^2 \rangle \end{bmatrix} \tag{1.16}$$

it is constituted by nine components, but considering its symmetry, the independent terms are six. Recalling equation (1.5) that defines the turbulent kinetic energy, it is possible to observe that the trace of the tensor corresponds to twice the turbulent kinetic energy k.

From a computational perspective, the grid requirements for RANS simulations are significantly less stringent than for DNS. The mesh must only be fine enough to resolve the gradients of the mean flow and the geometry, rather than the smallest Kolmogorov scales. Consequently, the computational cost does not scale theoretically with the Reynolds number as strictly as in DNS, making RANS the industry standard for high-Reynolds number engineering applications [4], with best performances for attached flows.

However, this reduction in computational effort comes at the cost of physical fidelity. By modeling the entire spectrum of turbulence, RANS simulations rely heavily on empirical constants and assumptions (e.g., isotropy of turbulence) that are not always accurate in complex flow regimes. Another fundamental limitation arises from the time-averaging procedure itself, that filters the entire dynamic content of the flow. Consequently, they are unable to capture transient phenomena, which are critical in many aerodynamic applications. While the mean flow representation is often sufficient for preliminary design, it fails when the physical unsteadiness of the large scales is dominant. To address cases where global flow instabilities must be resolved, the Unsteady RANS (URANS) approach can be adopted.

While those characteristics are generally true, it should be noted that the advantages and limitations of RANS simulations are significantly dependent on the turbulence model of choice. To have an overall understanding of their range of applicability, the main properties of those models will be summarized below.

To resolve the closure problem, varying levels of approximation can be adopted. Those approaches can be classified in the first place based on the order of resolved and modeled components (statistical moments).

First-order closures

The most common and simple class of approaches is represented by first-order closures. In this case the Reynolds stress tensor (that is a second moment), is determined through a relation with the mean flow velocity gradients (first moment). First-order closures can

be further classified in linear, mostly based on the Boussinesq hypothesis, and non-linear eddy-viscosity formulations.

In linear formulations, the modeling of the Reynolds stress tensor can be split in two contributions: a isotropic one, related to k, and a deviatoric one. To model the latter, the Boussinesq hypothesis is introduced. This assumption relies on a direct analogy with a laminar velocity field, assuming that the turbulent shear stresses are linearly proportional to the mean strain rate tensor E_{ij} through a scalar coefficient known as the eddy viscosity, ν_T:

$$-\langle u_i' \, u_j' \rangle = -\frac{2}{3} \, k \, \delta_{ij} + 2 \, \nu_T \, E_{ij} \tag{1.17}$$

The fundamental advantage of this hypothesis is the significant reduction in the problem complexity. It transforms the task of computing six independent tensor components into the determination of a single scalar field, ν_T, that represent a function of the fluid local state of motion. However this approach introduces significant limitations. The assumption of local equilibrium between the mean strain tensor E_{ij} and the deviatoric part of the turbulent stress tensor and its consequent independence on fluid rigid rotations is often excessively restrictive for flows characterized by massive separation, strong curvature, or unsteady large-scale structures [4] [5].

With the development of more complex non-linear eddy viscosity models, the limitations discussed above are in part mitigated. They extend the relation between the Reynolds stress tensor and E_{ij}, involving the mean flow rotation tensor Ω_{ij} and including also non-linear quadratic and cubic terms [6].

First-order turbulence closures can be also classified by the number of governing equations used, those with a greater number of equation usually provide a fuller description of turbulence.

- Algebraic or zero-equations models such as the uniform turbulent viscosity one are characterized by extremely limited ranges of applicability.
- One-equation models are also very limited in possible application fields, an example is the Spalart-Allmaras model, designed for aerodynamic flows. The only real advantages of using one-equation model rather than a two-equation model are related with stability and simplicity of the numerical solution [5].
- Two-equation models such as $k - \mathcal{E}$, $k - \omega$ and their blended implementations like Shear Stress Transfer (SST) are the most widely employed and for their importance will be further analyzed below.

Model $k - \mathcal{E}$ is based on the evaluation of the turbulent kinetic energy and dissipation rate. One of the main limitations is related to near wall behavior [4].

- In low Reynolds number turbulence models there is the need of some damping functions to limit k and $\mathcal{E}$ and an high grid resolution (first node at $y+ \leq 1$) in order to stay valid and resolve the viscous sublayer.

- By contrast, high Reynolds number turbulence models don't need damping functions, they are less stringent in term of grid resolution (first node inside $10 \leq y+ \leq 100$) but require the implementation of wall functions, often based in the logarithmic wall-law, with the consequent loss of accuracy in case of a separated flow.

Since the method is based on the Boussinesq assumption, it inherits all the limitations related to the linear eddy viscosity modeling already discussed previously. A third aspect regards the (non) universality: the model requires *ad hoc* adjustments of its calibration constants based on the nature of the considered flow [1]. It also degrade in performance with adverse pressure gradients.

Model $\boldsymbol{k-\omega}$ is the second most widely used two-equation model. The term ω is simply defined as: $\omega = \mathcal{E}/k$. It better performs for boundary layer flows and in presence of moderate adverse pressure gradients. The main drawback instead, is related to its sensibility to the arbitrary choice of ω boundary condition value [1] [5].

Model SST (Shear Stress Transport) merges the $k-\omega$ model with a high Reynolds number $k-\mathcal{E}$ model, with the intent of combining the advantages of the two. The $k-\omega$ is used in the boundary layer since it doesn't require damping functions and it better performs in presence of adverse pressure gradients, instead $k-\mathcal{E}$ is employed in wake regions of the boundary layer. The switching between the models is ensured by a blending function, dependent on wall proximity.

Second-order closures

Second-order closures, unlike first-order models that rely on the Boussinesq hypothesis represent a more complex and flexible class of approaches to determine the Reynolds stress tensor. Reynolds Stress Transport (RST) models allow to solve individual transport equations for the six independent component of the Reynolds stress tensor. This approach naturally include effects of streamline curvature, sudden changes in strain rate, secondary motions and the history effects [5]. With a similar accuracy, but a slightly lower level of description and complexity, the Algebraic Reynolds Stress (ARS) models employ only two transport equations, generally for the turbulent kinetic energy and for the dissipation rate. Despite their increased accuracy both RST and ARS models suffer numerical issues and they are more computationally demanding, so first-order models are more widely used in practice [4].

1.3.3 Large Eddy Simulation

The Large Eddy Simulation (LES) represents an intermediate approach between the high-fidelity of DNS and the low computational cost of RANS. The idea behind this technique is based on the observation that turbulent flows contain a wide range of length and time scales, which play different roles in the flow dynamics.

The large-scale structures, often referred to as "eddies", contain the majority of the turbulent kinetic energy and are responsible for the transport of momentum and conserved properties [4]. These scales are strictly dependent on the boundary conditions and the flow geometry, making them highly anisotropic and difficult to model. Conversely, the small-scale structures tend to be more isotropic and universal in their behavior, essentially acting as a sink for the kinetic energy transferred from the larger scales (energy cascade). Based on this distinction, LES applies a spatial filtering operation to the Navier-Stokes equations. The large, energy-carrying motions are explicitly resolved on the computational grid, while the effects of the unresolved scales, those smaller than the grid cut-off width, are modeled using Subgrid-Scale (SGS) models.

This strategy offers significant advantages over RANS approaches. Since the large-scale unsteady motions are resolved directly rather than averaged, LES is capable of capturing transient phenomena and flow instabilities, which are critical in aerodynamic flows involving separation or bluff bodies. The arbitrary aspects are confined only to the small scales, making the simulation less dependent on model calibration compared to RANS, where the entire turbulent spectrum is modeled.

However, these advantages come with a substantial computational cost. In terms of grid resolution requirements, LES places itself in an intermediate position between DNS and RANS, only requiring the resolution of the structures down to the cut-off width Δ. Consequently, the number of degrees of freedom is significantly lower than that of DNS, and substantially higher than RANS, where the mesh is only required to resolve the mean flow gradients. This requirement becomes particularly critical in wall-bounded flows, where the physical size of the energy-carrying eddies decreases proportionally to the distance from the wall. In the so-called Wall-Resolved LES (WR-LES), the need to resolve these inner boundary layer structures leads to a computational cost that scales significantly with the Reynolds number ($Re^{13/7}$), with possible prohibitive costs. To overcome this limitation and apply LES to high-Reynolds number flows of engineering interest, Wall-Modeled LES (WM-LES) approaches are adopted to bypass the strict resolution requirements of the near-wall region.

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